The internal architecture of $VO^+(4,2)$ and finite twistor geometry

The affine polar graph $VO^+(4,2)$ decomposes as the **tensor product $K_4 \otimes K_4$**, a result that flows directly from the hyperbolic quadric $Q^+(3,2) \cong PG(1,2) \times PG(1,2)$ underlying the determinant form on 2×2 matrices over GF(2). This tensor product structure, dual to the Cartesian product $K_4 \square K_4$ of the complement L2(4), generates all the graph's substructures: exactly **24 maximal cliques** (each of size 4, from the 6 totally singular 2-subspaces and their cosets), **8 maximum independent sets** (from 2 anisotropic 2-subspaces and their cosets), and a formally self-dual 2-class association scheme where the first eigenmatrix equals the second. Most strikingly, one of the two anisotropic subspaces defining the grid structure is **literally GF(4)** embedded in Mat(2,GF(2)) via the left regular representation — the companion matrix $\omega = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$ satisfies $\omega^2 + \omega + 1 = 0$. (Academic Kids) This GF(4) copy creates a direct algebraic bridge between the F_2 and F_4 Penrose constructions, and the 16 spacetime points embed into the Klein quadric $Q^+(5,2)$ as the big Schubert cell of $Gr_2(GF(2)^4)$, the finite analogue of affine complexified Minkowski spacetime. (Finitegeometry)

The tensor product decomposition and its consequences

$VO^+(4,2)$ is defined on the 16 elements of Mat(2×2, GF(2)), with adjacency when the difference matrix is singular (det = 0, rank ≤ 1). It carries parameters **srg(16, 9, 4, 6)** with eigenvalues 9^1, 1^9, $(-3)^6$. Its complement L2(4) = H(2,4) is the **4×4 rook graph** with parameters srg(16, 6, 2, 2) (Wikipedia +2) and eigenvalues 6^1, 2^6, $(-2)^9$.

The fundamental structural fact is that $VO^+(4,2) \cong K_4 \otimes K_4$, the tensor (categorical/Kronecker) product of two complete graphs on 4 vertices. (Wikipedia) In this product, vertices are pairs (i,j) with i,j ∈ {0,1,2,3}, and two vertices are adjacent if and only if they differ in **both** coordinates. This yields exactly 3 × 3 = **9 neighbors** per vertex. The common-neighbor counts follow immediately: two adjacent vertices (differing in both coordinates) share 2 × 2 = **4** common neighbors (λ = 4), and two non-adjacent vertices (sharing one coordinate) share 3 × 2 = **6** common neighbors (μ = 6). The complement $K_4 \square K_4$ is the Cartesian product, where adjacency means sharing **at least one** coordinate — precisely the rook's-move condition. (Tue) (Wikipedia)

The geometric origin of this product structure lies in the **ruled quadric** $Q^+(3,2) \cong PG(1,2) \times PG(1,2)$ in projective 3-space PG(3,2). The quadratic form Q(a,b,c,d) = ad + bc on $GF(2)^4$ defines a hyperbolic quadric with **9 isotropic points** (the rank-1 matrices) and **6 anisotropic points** (the invertible matrices $GL(2,2) \cong S_3$). The quadric has **two rulings** of 3 totally singular lines each, corresponding to the two PG(1,2) factors, and this bipartite ruling structure is what induces the tensor product decomposition.

Maximal cliques: 24 cliques of size 4

Every maximal clique of $VO^+(4,2)$ has size exactly 4, as the **Hoffman bound** gives $\omega \le 1 + k/(-s) = 1 + 9/3 = 4$. These cliques arise from the **6 totally singular 2-dimensional subspaces** of $(GF(2)^4, Q)$ — the 6 lines on the projective quadric $Q^+(3,2)$. The two rulings organize these as:

- **Ruling α (3 subspaces):** Matrices with zero second row {0, (1 0; 0 0), (0 1; 0 0), (1 1; 0 0)}, matrices with zero first row, and matrices with equal rows.
- **Ruling β (3 subspaces):** Matrices with zero second column, matrices with zero first column, and matrices with equal columns.

Each of the 6 subspaces has **4 cosets** in $GF(2)^4$, and every coset is a maximal clique. This gives 6 × 4 = **24 maximal cliques** total. Equivalently, in the $L2(4) = K_4 \square K_4$ picture, these 24 cliques correspond to the **24 transversals** (permutation patterns) of the 4×4 grid — one from each row and each column — matching the $|S_4| = 24$ permutations. Each vertex belongs to exactly **6 maximal cliques**, and each edge lies in exactly **2 maximal cliques**. Two totally singular subspaces in the same ruling are disjoint (as projective lines on $Q^+(3,2)$); two from different rulings meet in exactly one point.

Independent sets and the grid partition

The maximum independent sets of $VO^+(4,2)$ have size 4 (matching the Hoffman independence bound $\alpha \leq v\cdot(-s)/(k-s) = 16\cdot 3/12 = 4$), and there are exactly **8** of them, organized into **two parallel classes of 4**. These arise from the **two anisotropic 2-dimensional subspaces** — subspaces where all nonzero vectors are invertible (Q = 1). The matrix form of the bilinear form $B(x,y) = ad' + a'd + bc' + b'c$ and the requirement that $B(v_1,v_2) = 1$ for all basis pairs constrains the invertible matrices into exactly two complementary subspaces:

- **V_1 = {0, I, A, D}** where $A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$, $D = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$, satisfying $A = D^2$, $D = A^2$, $A + D = I$. This is the **cyclic subgroup** $\{e, \omega, \omega^2\}$ of $GL(2,2) \cong S_3$ (the "rotations"/even permutations) plus zero.
- **V_2 = {0, J, B, C}** where $J = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$, $C = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. These are the **three involutions** of GL(2,2) (the "reflections"/odd permutations) plus zero.

These satisfy $V_1 \perp V_2$ under both the bilinear form B and the standard inner product, $V_1 \oplus V_2 = GF(2)^4$, and $V_2 = J \cdot V_1 = V_1 \cdot J$. The **4 cosets of V_1** give the "rows" of the L2(4) grid; the **4 cosets of V_2** give the "columns." Every row meets every column in exactly one matrix, establishing the unique $K_4 \square K_4$ decomposition of L2(4). In terms of matrix entries $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, the row coordinate is **(a+b+d, b+c)** and the column coordinate is **(a+b+c, a+d)**, both elements of $GF(2)^2$.

The formally self-dual association scheme

The 16 matrices form a **2-class symmetric association scheme** (3 relations including identity), stratified by the rank of the difference: rank 0 (identity, valency 1), rank 1 (9 neighbors, giving $VO^+(4,2)$), and rank 2 (6 neighbors, giving L2(4)).

Eigenmatrices and self-duality

The first eigenmatrix (P-matrix) is:

Eigenspace	Multiplicity	A_0	A_1 (rank 1)	A_2 (rank 2)
E_0	1	1	9	6
E_1	9	1	1	−2
E_2	6	1	−3	2

The scheme is **formally self-dual**: **P = Q**. Since $P^2 = 16I = nI$, the second eigenmatrix $Q = (n/1)P^{-1} = P$. The Krein parameters therefore equal the intersection numbers: **q^k_{ij} = p^k_{ij}** for all i,j,k. All Krein parameters are non-negative integers, a strong structural property. This self-duality is realized concretely through the **trace pairing** $tr(M^TN)$ on Mat(2×2, GF(2)), which identifies the dual group with the original group. As a translation scheme on $(Z/2)^4$, the characters χ_N(M) = (−1)^{tr(N^TM)} partition into eigenspaces indexed by rank(N), mirroring the original rank stratification University of Toronto — the hallmark of strong self-duality in bilinear forms schemes (following Delsarte, 1978). Uni-paderborn

Intersection numbers

The complete intersection algebra is:

$\mathbf{A_1^2 = 9A_0 + 4A_1 + 6A_2}$ and $\mathbf{A_2^2 = 6A_0 + 2A_1 + 2A_2}$ and $\mathbf{A_1A_2 = 4A_1 + 3A_2}$

Key intersection numbers: for two rank-1-adjacent matrices (det(M−N) = 0), there are $\mathbf{p^1_{11} = 4}$ common rank-1 neighbors and $\mathbf{p^1_{22} = 2}$ common rank-2 neighbors. For two rank-2-adjacent matrices (det(M−N) = 1), there are $\mathbf{p^2_{11} = 6}$ common rank-1 neighbors and $\mathbf{p^2_{22} = 2}$ common rank-2 neighbors. The scheme is both **P-polynomial** and **Q-polynomial** with classical parameters (D,b,α,β) = (2,2,1,3).

Fusion schemes

For a 2-class scheme, the only non-trivial partition of $\{R_1, R_2\}$ is $\{R_1 \cup R_2\} \to K_{16}$ (trivial). Each individual class already defines a strongly regular graph ($VO^+(4,2)$ and L2(4)), but no additional non-trivial fusion schemes exist. The minimal idempotents are $E_0 = (1/16)J$, $E_1 = (1/16)(9I + A_1 - 3A_2)$ of rank 9, and $E_2 = (1/16)(6I - 2A_1 + 2A_2)$ of rank 6.

GF(4) as the algebraic bridge between constructions

The subspace $V_1 = \{0, I, A, D\}$ is not merely a vector subspace — it is **isomorphic to GF(4) as a ring** under matrix addition and multiplication. The element $D = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$ is a **primitive cube root of unity** satisfying $D^2 + D + I = 0$, with $D^3 = I$. This is precisely the **companion matrix** of the irreducible polynomial $x^2 + x + 1$ over GF(2), and V_1 realizes the **left regular representation** of GF(4) over GF(2). Every element of GF(4) = $\{0, 1, \omega, \omega^2\}$ maps to its multiplication matrix: Academic Kids $0 \mapsto 0$, $1 \mapsto I$, $\omega \mapsto D$, $\omega^2 \mapsto A$.

This identification has profound implications for the relationship between the F_2 and F_4 Penrose constructions:

The grid structure of L2(4) encodes the GF(4) arithmetic. Two matrices lie in the same "row" of the $K_4 \square K_4$ grid if and only if their difference belongs to $GF(4) \subset Mat(2,GF(2))$. The row partition is precisely the **coset decomposition** $Mat(2,GF(2)) / GF(4)$. Two matrices lie in the same "column" if and only if their difference belongs to $V_2 = J \cdot GF(4)$, the "conjugate copy" of GF(4) obtained by left multiplication by the permutation matrix $J = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

The Frobenius automorphism $\sigma: x \mapsto x^2$ of GF(4)/GF(2) acts on V_1 by swapping $D \leftrightarrow A$ (i.e., $\omega \leftrightarrow \omega^2$). In the matrix representation, σ is realized as **conjugation by $P = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$**: $P^{-1}DP = A$ and $P^{-1}AP = D$. Since $P \in V_2$, the Frobenius automorphism exchanges the two non-trivial elements of GF(4)* while preserving V_1 as a set. This same P encodes the Frobenius on the ambient finite geometry.

How 16 spacetime points sit inside the Klein quadric

The big Schubert cell embedding

The 35 lines of PG(3,2) map bijectively to the **35 points of $Q^+(5,2)$** in PG(5,2) via the Plücker embedding. (Finitegeometry) (Prespacetime) The 16 spacetime matrices embed as a specific 16-element subset of these 35 points — the **big Schubert cell** of the Grassmannian $Gr_2(GF(2)^4)$.

Fix the decomposition $GF(2)^4 = V_1 \oplus V_2$ (where $V_1 = GF(4)$ and $V_2 = J \cdot GF(4)$). Each 2×2 matrix $M = \begin{pmatrix} a & c \\ b & d \end{pmatrix}$ determines the 2-plane that is the row space of the 4×2 block $[I_2; M]^T$ — equivalently, the graph of the linear map $V_2 \to V_1$ given by M. The **16 matrices parametrize exactly those 2-planes complementary to V_1** (intersecting V_1 trivially). Their Plücker coordinates are:

$$(p_{01}, p_{02}, p_{03}, p_{12}, p_{13}, p_{23}) = (1, c, d, a, b, ad+bc)$$

All 16 points satisfy the Klein quadratic relation $p_{01}p_{23} + p_{02}p_{13} + p_{03}p_{12} = (ad+bc) + cb + da = 2(ad+bc) = 0$ over GF(2), confirming they lie on $Q^+(5,2)$. (Finitegeometry) (arXiv) The sixth coordinate $ad+bc = \det(M)$ is the quadratic form itself, encoding the null/non-null distinction directly in the Plücker embedding.

Important correction: The 16 spacetime points are not "one family of maximal planes" on $Q^+(5,2)$. Each family of generators consists of **15 planes** (corresponding to the 15 points or 15 planes of PG(3,2)), (Finitegeometry) not 16. (Finitegeometry) (ResearchGate) The 16 matrices form an **affine substructure** — the finite analogue of affine complexified Minkowski space C^4 sitting inside the compact Grassmannian $Gr_2(C^4)$.

The 35 = 16 + 18 + 1 decomposition

The remaining 19 lines of PG(3,2) — those not in the big Schubert cell — decompose by their intersection with V_1:

- **16 points** ($\dim(W \cap V_1) = 0$): The big cell, affine spacetime, parametrized by Mat(2×2, GF(2))

- **18 points** ($\dim(W \cap V_1) = 1$): Lines meeting V_1 in a 1-space, organized as **3 pencils of 6** (one pencil for each of the 3 points of $PG(V_1) \cong PG(1,2)$)
- **1 point** ($\dim(W \cap V_1) = 2$): The line V_1 itself — the "point at infinity" corresponding to the chosen reference 2-plane

The 18 + 1 = 19 boundary points form the "conformal completion" of the 16-point affine spacetime.

The 28 off-quadric points and E_7

The 28 points of PG(5,2) not on $Q^+(5,2)$ form the **triangular graph $T(8) = L(K_8)$**, (MDPI) with parameters srg(28, 12, 6, 4). (Wikipedia) (Frans Marcelis) These 28 points are labeled by 2-element subsets {i,j} of an 8-set, (Frans Marcelis) with adjacency when two pairs share an element. (Wikipedia) The connection to the **E_7 root system** runs through the isomorphism $W(E_7)/\{\pm 1\} \cong Sp(6,F_2)$, which is the automorphism group of $Q^+(5,2)$. The 28 off-quadric points correspond to the **28 positive roots of E_7** (equivalently, the 28 bitangents of a quartic curve). (Frans Marcelis) They decompose into **8 Conwell heptads** of 7 points each (M759) (Finitegeometry) — any two heptads share exactly 1 point, and each point belongs to exactly 2 heptads (MDPI) — reflecting the S_8 symmetry and the foundational isomorphism **$PSL(4,2) \cong A_8$** (M759) (qmul) established by Conwell (1910).

Saniga (2015) proved that the $(28_6, 56_3)$-configuration of 28 off-quadric points and 56 external lines is isomorphic to the **combinatorial Grassmannian $G_2(8)$**, whose points are 2-subsets and lines are 3-subsets of {1,…,8}. (MDPI)

The $F_2 \to F_4$ commutative diagram

Baer subgeometry embedding

PG(3,2) embeds into PG(3,4) as a **Baer subgeometry** — precisely the **fixed-point set of the Frobenius automorphism** σ: $(x_0:x_1:x_2:x_3) \mapsto (x_0^2:x_1^2:x_2^2:x_3^2)$. The 15 F_2-rational points (homogeneous coordinates in $\{0,1\}^4$) sit inside the 85 points of PG(3,4). Each line of PG(3,2) (with 3 points) extends to a line of PG(3,4) (with 5 points), forming a **Baer subline**. At the spacetime level, $Mat(2\times 2, F_2) \cong F_2^4$ embeds as an additive subgroup of index 16 in $Mat(2\times 2, F_4) \cong F_4^4$. (PubMed Central) The determinant polynomial has F_2-coefficients, so the null separation graph on $Mat(2\times 2, F_2)$ is the **induced subgraph** of the null separation graph on $Mat(2\times 2, F_4)$.

The Hermitian form and its restriction

Over F_4, the Hermitian form $H(x,y) = \Sigma\, x_i \cdot y_i^2$ defines the **Hermitian variety H(3,4)** (HandWiki) in PG(3,4), a generalized quadrangle **GQ(4,2)** (Ugent) with **45 isotropic points** (null twistors) and **40 non-isotropic points** among the 85 total. The associated norm $Q(x) = H(x,x) = \Sigma\, x_i^3$ maps to GF(2) via the norm map $N: GF(4) \to GF(2)$.

When H restricts to F_2^4, the conjugation σ becomes the identity (since $a^2 = a$ for $a \in F_2$), and the Hermitian form degenerates to the **standard dot product** $B(x,y) = \Sigma\, x_i y_i$ over F_2. The associated quadratic form becomes Q(x)

$= \Sigma x_i$, a linear form. The intersection **H(3,4) ∩ PG(3,2)** is the hyperplane $\{\Sigma x_i = 0\}$, which is a **copy of PG(2,2) — the Fano plane — with 7 points**. So the 15 F_2-rational twistor points split as **7 isotropic** (on H(3,4)) and **8 non-isotropic**.

Generalized quadrangle hierarchy

The restriction creates a natural embedding chain of generalized quadrangles:

The **symplectic generalized quadrangle W(2) = GQ(2,2)** with 15 points and 15 lines (the "doily") embeds into **H(3,4) = GQ(4,2)** with 45 points and 27 lines. W(2) is the F_2 structure; GQ(4,2) is the F_4 structure. The embedding is the finite geometric content of the field extension $F_2 \to F_4$. Explicitly, the 15 isotropic lines of PG(3,2) (under the symplectic form from restricting H) form W(2), while the full 45-point Hermitian variety over F_4 gives the larger GQ(4,2), which is dual to the elliptic quadric Q(5,2). ugent Tue

The commutative diagram

The **Klein correspondence commutes with the field extension**, yielding:

```
PG(3,2)  ——Baer subgeometry——→  PG(3,4)
  |                              |
  | Klein correspondence         | Klein correspondence
  ↓                              ↓
Q⁺(5,2) ⊂ PG(5,2) ————————————→ Q⁺(5,4) ⊂ PG(5,4)
```

The Frobenius σ acts on each space, fixing the F_2-rational substructure. Plücker coordinates transform covariantly under σ, so the correspondence is equivariant. The double fibration that underlies the Penrose transform — the incidence correspondence F(1,2; V) parametrizing point-line flags — embeds compatibly:

```
      F(1,2; F₂⁴)  ———→  F(1,2; F₄⁴)
     /        \        /        \
    /          \      /          \
PG(3,2)      Q⁺(5,2)  PG(3,4)    Q⁺(5,4)
(twistors)  (spacetime)  (twistors) (spacetime)
```

Over finite fields, the "Penrose transform" becomes a **combinatorial incidence correspondence** rather than a cohomological one. Lévay and Holweck (Phys. Rev. D 99, 2019) formalized this as a finite geometric model of bulk-boundary correspondence, where PG(3,2) is the "boundary" (twistor space) and $Q^+(5,2)$ is the "bulk" (spacetime), with the structure of a **geometric subspace error-correcting code**. arXiv

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Natural quotient structures and the role of GF(4)

Quotients by independent sets yield K_4

Partitioning the 16 vertices into the 4 rows (cosets of $V_1 = GF(4)$) gives an **equitable partition** where each vertex has exactly **3 neighbors in each of the other 3 blocks** and **0 neighbors in its own block**. The quotient matrix is $3(J_4 - I_4)$ with eigenvalues 9 and −3, both eigenvalues of $VO^+(4,2)$. The **quotient graph is K_4** with uniform edge multiplicity 3. An identical result holds for the column partition (cosets of $V_2 = J \cdot GF(4)$). Between any two distinct blocks, there are $4 \times 3 =$ **12 directed edges** (or 12 edges counting both endpoints).

Quotients by cliques yield K_4 with loops

Partitioning by 4 cosets of a totally singular 2-subspace gives cliques. Within each block: **3 edges** (K_4). Between blocks: each vertex has **2 neighbors** per other block. The quotient has diagonal entries 3 and off-diagonal entries 2, with eigenvalues 9 and 1.

Why two quotient types exist

The two quotient types correspond to the **two types of 2-dimensional subspaces**: anisotropic (all nonzero elements invertible → independent sets → quotient K_4) and totally singular (all nonzero elements singular → cliques → quotient K_4 with loops). There are exactly **2 anisotropic subspaces** (V_1 and V_2) and **6 totally singular subspaces** (the lines on $Q^+(3,2)$).

The critical tensor product identity

The perpendicularity $V_1 \perp V_2$ under the bilinear form B produces a remarkable identity: for any $v_1 \in V_1 \setminus \{0\}$ and $v_2 \in V_2 \setminus \{0\}$, we have $Q(v_1 + v_2) = Q(v_1) + Q(v_2) + B(v_1,v_2) = 1 + 1 + 0 = 0$ over GF(2). Thus **every sum of a nonzero element from each anisotropic subspace is singular**. Combined with the fact that V_1 and V_2 are themselves entirely non-singular, this yields the tensor product structure: two matrices from different rows AND different columns always differ by a singular matrix (adjacent in VO^+), while matrices sharing a row or column always differ by an invertible matrix (non-adjacent). This is precisely the adjacency rule of $\mathbf{K_4 \otimes K_4}$.

The 10 singular matrices decompose as $\{0\} \cup \{v_1 + v_2 : v_1 \in V_1\{0\}, v_2 \in V_2\{0\}\}$, giving $1 + 3\times3 = 10$. The 6 invertible matrices are $(V_1\{0\}) \cup (V_2\{0\}) = \{I,A,D\} \cup \{J,B,C\}$, the "rotations" and "reflections" of $S_3 \cong GL(2,2)$.

Conclusion: emergent structure and unifying themes

Three structural insights emerge from this analysis that are likely significant for the finite Penrose transform program.

First, GF(4) is intrinsic to the F_2 construction. The grid decomposition of $L2(4) = K_4 \square K_4$ forces the appearance of GF(4) as one of the two anisotropic subspaces — not as an external extension but as an internal

substructure of Mat(2,GF(2)). The "rows" of the spacetime grid are precisely cosets of this GF(4) copy. This means the F_4 construction is already latent in the F_2 geometry: extending scalars from F_2 to F_4 amounts to "completing" the row coordinate from a coset label to a full field element.

Second, the formally self-dual association scheme creates a perfect duality between spacetime and frequency space. The identity P = Q means that the rank-stratification of Mat(2,GF(2)) is invariant under the Fourier transform (character expansion) on $(Z/2)^4$. This self-duality, realized through the trace pairing, is a finite analogue of the conformal invariance that makes the classical Penrose transform work — the symmetry between position-space and momentum-space descriptions. ScienceDirect

Third, the Schubert cell embedding $Q^+(5,2) \supset AG(4,2)$ connects spacetime, the Klein quadric, and E_7 in a single 63-point projective space. The 63 points of PG(5,2) decompose as 16 (affine spacetime) + 19 (conformal boundary) + 28 (off-quadric / E_7 roots), with the 35 = 16 + 19 on-quadric points forming the compactified spacetime and the 28 off-quadric points carrying the E_7 root structure via T(8). Frans Marcelis The generalized quadrangle hierarchy $W(2) \subset GQ(4,2) = H(3,4)$ provides the geometric mechanism through which the F_2 twistor structure embeds Frans Marcelis Frobenius-equivariantly into the F_4 twistor structure, with the Klein correspondence commuting with the field extension at every level.